METRIC

DOD-P-70330
16 August 1982

MILITARY SPECIFICATION

PLUG, EXPANSION, METRIC,
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metrically dimensioned expansion plug (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
QQ-Z-325 - Zinc Coating, (Electrodeposited), Requirements for
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

MILITARY

DOD-P-16232 - Phosphate Coating, Heavy, Manganese or Zinc Base (for Ferrous Metals)
DOD-P-70330/1 - Plug, Expansion, Carbon Steel, Metric

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

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MILITARY

DOD-P-70330/2 - Plug, Expansion, CRES, Metric
DOD-P-70330/3 - Plug, Expansion, Aluminum Alloy, Metric

STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-1312 - Fasteners, Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AEROSPACE MATERIAL SPECIFICATION (AMS)

AMS 2470 - Anodic Treatment of Aluminum Alloys, Chromic Acid Process
AMS 4003 - Aluminum, Sheet and Plate (1100-H14)
AMS 5510 - Steel Sheet, Strip and Plate, Corrosion and Heat Resistant, 18 Cr - 10.5 Ni - 0.40 Ti (SAE 30321)

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ANSI/ASTM A 380 - Cleaning and Descaling Stainless Steel Parts, Equipment and Systems

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references, other than specification sheets, cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable. The expansion plugs shall be fabricated of the following materials as specified.

3.2.1 Carbon steel. Carbon Steel shall be composition 1008 thru 1022 (UNS G10080 thru G10220) in accordance with FED-STD-66.

3.2.2 Corrosion resistant steel. Corrosion resistant steel shall be in accordance with AMS 5510 (UNS S32100).

3.2.3 Aluminum alloy. Aluminum alloy shall be in accordance with AMS 4003 (UNS A91100).

3.3 Surface hardness. Expansion plugs fabricated of carbon steel shall exhibit a hardness of 50-80HRB when tested as specified in 4.6.2.

3.4 Protective finish.

3.4.1 Carbon steel. Expansion plugs fabricated of carbon steel shall be protected wth one of the following finishes as specified (see 3.1).

3.4.1.1 Phosphate coating. Phosphate coating shall be in accordance with DOD-P-16232, type M, class 3.

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3.4.1.2 Cadmium plate. Cadmium plate shall be in accordance with QQ-P-416, type II, class 2 (7.6 µm thick).

3.4.2 Cleaning and descaling. Corrosion resistant steel plugs shall be cleaned and descaled in accordance with ASTM A380.

3.4.3 Aluminum alloy. Expansion plugs fabricated of aluminum alloy shall be subjected to an anodic treatment in accordance with AMS 2470.

3.5 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet and shall apply after protective finishes.

3.5.1 Edges. All edges shall be broken, 0.127 mm, minimum.

3.6 Surface texture. Plugs shall have a maximum surface texture of 3.2 µm, when tested as specified in 4.6.3.

3.7 Cracks. Plugs shall be free from cracks, flaws and pits in any location when tested as specified in 4.6.4.

3.8 Identification marking. The plug shall be identified in accordance with MIL-STD-130.

3.9 Workmanship. Plugs shall be free from surface contamination, tool marks and other imperfections which may adversely affect useability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.1, 4.6.2 and 4.6.3.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the plugs are in accordance with the requirements of 3.2 and 3.3.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table I.

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Group A		
Examination	1/	4.6.1
Cracks	3.7	4.6.4
Group B		
Surface Hardness	3.3	4.6.2
Surface Texture	3.6	4.6.3

1/ All requirements not covered by tests.

4.5.1 Inspection lot. An inspection lot shall consist of all plugs covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.5.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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4.5.3 Sampling for group A inspection. Sampling shall be in accordance with MIL-STD-105 inspection level III with an Acceptable Quality Level (AQL) of 0.25 percent defective for major defects. For minor defects inspection level I with an AQL of 1.0 percent defective shall apply.

4.5.4 Sampling for group B inspection. Sampling shall be in accordance with MIL-STD-105 inspection level S-2 with an AQL of 4.0 percent defective and may be made on sample units which have been subjected to group A inspection. Different sample units shall be selected for each group B test.

4.5.5 Classification of defects. Defects shall be classified as specified in table II.

TABLE II. Classification of defects.

Category	Defect
Major	
101	Diameter
102	Thickness
103	Marred or dull edge on concave side
104	Cracks
Minor	
201	Curvature (sphericity)
202	Sides parallel
203	Coating of dirt, oil or other foreign substance
204	Pits, rust, loose scale, scrapes, splits, cracks
205	Other injurious defects
206	Incorrect marking identification
207	Workmanship not as specified
208	Packaging not as specified
209	Other dimensional characteristics not covered above

4.5.6 Packaging inspection. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with PPP-H-1581.

4.6 Methods of inspection.

4.6.1 Examination. Sample plugs shall be thoroughly examined to determine conformance with this specification and applicable specification sheets with respect to all the requirements not covered by tests.

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4.6.2 Surface hardness. Sample plugs shall be tested for surface hardness in accordance with MIL-STD-1312 test method no. 6.

4.6.3 Surface texture. Sample plugs shall be tested for surface texture in accordance with ANSI B46.1.

4.6.4 Cracks. Sample plugs shall be examined for cracks with 10X magnification. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign matter.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Expansion plugs acquired under this specification are intended for closng core holes in castings and other similar openings in castings.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.
b. Applicable Military specification sheet part number(s).
c. Applicable levels of packaging protection (see 5.1).

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AT, AV, EA, ER
DLA - IS

User activities:
Army - ME
Navy - MC

Agent:
DLA - IS

Preparing activity:
Army - AR

(Project 5340-1345)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE DOD-P-70330
PLUG, EXPANSION, METRIC, GENERAL SPECIFICATION FOR

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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